

Comprehensive Remote Server & Dataset Manual

Remote Workflow Documentation

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1 Introduction

This manual guides you through managing long-running Python scripts on a remote Linux server. Since you are learning these steps, this guide explains not just *what* to type, but *why* you are typing it.

2 Phase 1: The Connection (SSH)

Concept: What is SSH?

SSH (Secure Shell) is a protocol that allows you to control a remote computer safely. Think of it as opening a "text-only" window into the remote server. When you type commands, they run on the server, not your laptop.

Step 1: Log in to the Server

Open your terminal and run the following command. You will be asked for a password (text will not appear while typing).

```
ssh avik@137.110.198.246
```

3 Phase 2: File System Access (Mounting)

Concept: Why do we "Mount" a drive?

On servers, extra hard drives (like NVMe SSDs) are not always automatically connected when the computer turns on. "Mounting" is the act of telling the operating system: *"Take this physical hardware (/dev/nvme...) and make it accessible at this folder path (/mnt/nvme...)."*

Step 2: Check & Mount the Drive

Scenario A: Routine Check

Before working, always check if your data folder exists and is accessible.

```
df -h | grep nvme  
# If you see a line ending in "/mnt/nvme_data", you are good to go.
```

Scenario B: The Drive is Missing (e.g., after a Reboot)

If the command above returns nothing, you must mount the drive manually.

```
# 1. Mount the physical partition to the folder
sudo mount /dev/nvme2n1p1 /mnt/nvme_data

# 2. Verify it worked
df -h /mnt/nvme_data
```

Step 3: Enter the Project Folder

Once mounted, navigate to your code directory:

```
cd /mnt/nvme_data/Avik_Shubhan_codes_data
```

4 Phase 3: The Persistent Environment (Tmux)

Concept: Why Tmux?

Normally, if you close your laptop or lose WiFi, your SSH connection breaks and your running scripts **die immediately**.

Tmux creates a "session" that lives on the server background. If you disconnect, the session keeps running. You can "detach" (leave it running) and "attach" (come back to it) later.

The "Golden Rule" of Tmux

Almost every command in tmux requires a **Prefix Key**. You must press:

Ctrl + B

Release those keys, and *then* press your desired command key.

Scenario 1: Starting a Fresh Task

Use this when you are starting a script for the first time.

```
# Syntax: tmux new -s [session_name]
tmux new -s dataset_build
```

You are now inside tmux. You will see a green bar at the bottom of the screen.

Scenario 2: Leaving Work for the Night (Detaching)

You want the script to keep running, but you want to close your laptop.

1. Press **Ctrl + B**.
2. Release keys.
3. Press **d**.

You will return to the normal command prompt. The script is still running in the background.

Scenario 3: Resuming Work (Attaching)

You logged back in and want to see your progress.

```
# 1. See what is running
tmux ls

# 2. Reconnect to the specific session
tmux attach -t dataset_build
```

Scenario 4: Cleaning Up

If a script is finished or messed up, and you want to delete the session entirely:

```
tmux kill-session -t dataset_build
```

5 Phase 4: Execution Workflow (Your Scripts)

This section combines everything into the actual tasks you need to perform.

Task A: Running Step 8 (Dataset Creation)

1. **SSH in:** `ssh avik@137.110.198.246`
2. **Mount:** (If needed) `sudo mount /dev/nvme2n1p1 /mnt/nvme_data`
3. **Go to folder:** `cd /mnt/nvme_data/Avik_Shubhan_codes_data`
4. **Start Tmux:** `tmux new -s dataset_build`
5. **Run Python:**

```
python3 Part\ 8.py
```

6. **Detach:** Ctrl+B, then d.

Task B: Running Step 9 (ResNet Training)

1. **SSH in:** `ssh avik@137.110.198.246`
2. **Go to folder:** `cd /mnt/nvme_data/Avik_Shubhan_codes_data`
3. **Start Tmux:** `tmux new -s resnet_dataset`
4. **Run Python:**

```
python3 Part\ 9_v3.py
```

5. **Detach:** Ctrl+B, then d.

6 Advanced Tmux: Windows & Panes

Sometimes you want to monitor CPU usage (htop) while your script runs, without opening a second SSH connection.

Using Windows (Like Browser Tabs)

- **New Window:** `Ctrl+B`, then `c`. (Creates a fresh shell 0, 1, 2...)
- **Next Window:** `Ctrl+B`, then `n`.
- **Previous Window:** `Ctrl+B`, then `p`.
- **List Windows:** `Ctrl+B`, then `w`. (Use arrows to select).

Using Panes (Split Screen)

- **Split Horizontal:** `Ctrl+B`, then `"` (Shift + Quote).
- **Split Vertical:** `Ctrl+B`, then `%` (Shift + 5).
- **Switch Panes:** `Ctrl+B`, then Arrow Keys.
- **Close Pane:** Type `exit` or press `Ctrl+d`.

Scrolling Up (Copy Mode)

You cannot scroll up with your mouse wheel in tmux by default.

1. Press `Ctrl+B`, then `[`.
2. You are now in "Copy Mode". A yellow indicator appears at the top right.
3. Use **Up/Down Arrows** or **PageUp/PageDown** to scroll history.
4. Press `q` to quit Copy Mode and return to typing.